Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 35)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain why introduced species often have a greater effect on island communities than on mainland communities.

[2 marks]

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0 1 . 2 For each of the following, describe an example of an introduced species and explain its effect on an indigenous species.

[6 marks]

Introduced competitor .....

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Introduced predator .....

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Introduced pathogen .....

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0 1 . 3 Explain how hybridisation with an introduced species threatens the red deer and the Scottish wildcat.

[2 marks]

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0 2 . 1 Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking .....

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DNA databases and eDNA .....

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Image recognition software .....

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Acoustic monitoring / sonograms .....

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0 2 . 2 Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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0 3 A student is comparing the plant diversity of a grazed and an ungrazed grassland, and estimating the population of grasshoppers in each. In the grazed field 60 grasshoppers were caught, marked and released; three days later 75 were caught, of which 15 were marked.

0 3 . 1 Explain **three** reasons for choosing quadrat positions using random numbers rather than by eye.

[3 marks]

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0 3 . 2 Use the Lincoln index to estimate the grasshopper population of the grazed field.

[2 marks]

Answer

0 3 . 3 State **three** assumptions that must hold for this estimate to be valid.

[3 marks]

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0 3 . 4 Explain how the student should decide the size of quadrat and the number of quadrats to use.

[3 marks]

0 3 . 5 Explain **four** other features of the design that would make the comparison between the two fields more reliable or more valid.

[4 marks]

15

0 4 . 1 Explain the difference between primary and secondary succession and why secondary succession is faster.

[3 marks]

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0 4 . 2 Explain what is meant by **deflected succession** and a **plagioclimax**, using a named UK example.

[3 marks]

0 4 . 3 Explain why conservation of a plagioclimax requires continued human activity whereas conservation of a rainforest or coral reef does not.

[2 marks]

0 4 . 4 Explain why biodiversity tends to be low in abiotically extreme environments and high in stable ones.

[2 marks]

10

0 5 A conservation organisation is converting 700 ha of arable farmland on the coast into a wetland reserve for breeding and wintering birds (as at Wallasea Island).

0 5 . 1 Explain why habitat area and shape are important in the design of the reserve.

[4 marks]

0 5 . 2 Explain how the reserve could be designed to provide a range of niches.

[3 marks]

0 5 . 3 Explain why some species may need to be deliberately introduced while others colonise naturally.

[2 marks]

0 5 . 4 Explain the value of a biological corridor linking the reserve to other wetlands.

[1 mark]

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0 6 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 6 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 7 Table 1 shows the energy inputs and food energy outputs of three farming systems, in GJ ha^{-1} yr^{-1}.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Subsistence maize, hand cultivated	0.6	9.0
Intensive wheat, UK	18.0	54.0
Intensive beef, feedlot	40.0	12.0

0 7 . 1 Calculate the energy ratio (output / input) for each system in Table 1.

[3 marks]

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Answer

0 7 . 2 Explain what is meant by an **energy subsidy**, and use your answers to explain why the intensive systems have much lower energy ratios.

[3 marks]

0 7 . 3 Explain **three** ways a farmer can reduce the energy losses in a food production chain.

[3 marks]

0 7 . 4 Explain **two** advantages of genetic modification over conventional selective breeding.

[2 marks]

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0 7 . 5 Explain **four** concerns about the use of GM crops.

[4 marks]

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0 8 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 8 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 8 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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0 9 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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2 .....

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0 9 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 9 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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1 0 . 1 Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

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1 0 . 2 State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity.

[25 marks]

OR

1 1 . 2 Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS